

Grade 12
Cycle Test Memo

25 March

Question 1

⊖1 Max for Q1 for notation error

1.1. $f(x) = -\frac{2}{x}$

$$f(x+h) = -\frac{2}{x+h}$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \quad \checkmark$$

$$= \lim_{h \rightarrow 0} \frac{-\frac{2}{x+h} - \left(-\frac{2}{x}\right)}{h} \quad \checkmark$$

$$= \lim_{h \rightarrow 0} \frac{-2x + 2(x+h)}{x(x+h)} \div h$$

$$= \lim_{h \rightarrow 0} \frac{-2x + 2x + 2h}{x^2 + xh} \div h$$

$$= \lim_{h \rightarrow 0} \frac{2h \checkmark}{x^2 + xh} \times \frac{1}{h}$$

$$= \lim_{h \rightarrow 0} \frac{2}{x^2 + xh}$$

$$= \frac{2}{x^2 + x(0)} \quad \checkmark$$

$$= \frac{2}{x^2} \quad \checkmark$$

(5)

$$1.2.1. \quad y = 16x^2 + \frac{3}{x^{10}}$$

$$y = 16x^2 + 3x^{-10} \checkmark A$$

$$\frac{dy}{dx} = 32x - 30x^{-11} \quad (3)$$

$$1.2.2. \quad D_x \left[x^2 (3x - \sqrt[3]{x^2}) \right]$$

$$D_x \left[x^2 (3x - x^{\frac{2}{3}}) \right]$$

$$D_x \left[3x^3 - x^{\frac{8}{3}} \right] \checkmark A$$

$$= 9x^2 - \frac{8}{3}x^{\frac{5}{3}} \quad (3)$$

$$1.2.3. \quad f(x) = \frac{8 - 27x^3}{2 - 3x}$$

$$f(x) = \frac{(2 - 3x)(4 + 6x + 9x^2)}{(2 - 3x)} \checkmark$$

$$f(x) = 4 + 6x + 9x^2$$

$$f'(x) = 6 + 18x \checkmark \quad (4)$$

Question 2

$$2.1. \quad f(x) = \overset{a=1}{1}(x+1)^2(x-1) \quad \checkmark x\text{-int}$$

$$= (x^2 + 2x + 1)(x - 1)$$

$$= x^3 + 2x^2 + x - x^2 - 2x - 1$$

$$= x^3 + x^2 - x - 1$$

$$\therefore p=1 \quad q=-1 \quad r=-1$$

(3)

$$2.2. \quad f'(x) = 3x^2 + 2x - 1 \quad \checkmark f'$$

$$3x^2 + 2x - 1 = 0 \quad \checkmark f' = 0$$

$$(3x - 1)(x + 1) =$$

$$x = \frac{1}{3} \quad \checkmark \quad \text{or} \quad x = -1$$

N/A

$$f\left(\frac{1}{3}\right) = -\frac{32}{27}$$

$$\therefore C\left(\frac{1}{3}; -\frac{32}{27}\right) \quad \checkmark \text{ Must be in coordinates}$$

(4)

$$2.3. \quad f'(z) = 3(z)^2 + 2(z) - 1$$

$$= 15 \quad \checkmark f'(z) = 15 \quad \therefore m = 15$$

$$f(z) = (z)^3 + (z)^2 - 2 - 1$$

$$= 9 \quad (z; 9) \checkmark^A$$

$$\therefore y - 9 = 15(x - 2)$$

$$y = 15x - 21 \quad \checkmark \quad (3)$$

$$2.4 \quad x^3 + x^2 - x - 1 = 15x - 21 \quad \checkmark^m$$

$$x^3 + x^2 - 16x + 20 = 0 \quad \checkmark^{SF}$$

$$(x - 2)(x^2 + 3x - 10) = 0$$

$$(x - 2)(x + 5)(x - 2) = 0 \quad \checkmark \text{Factors}$$

$$\therefore x = 2 \quad \text{or} \quad \underline{x = -5} \quad \checkmark \quad (4)$$

N/A

$$2.5 \quad x < -\frac{1}{3} \quad \checkmark \quad (2)$$

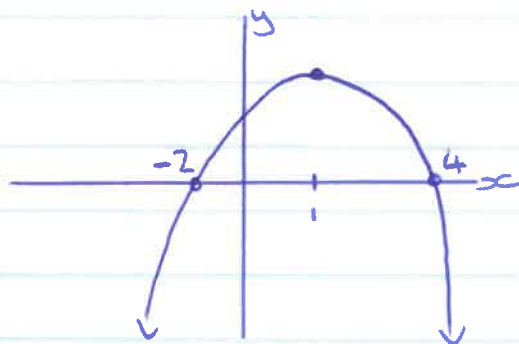
Alternate: $f''(x) = 6x + 2$

$$6x + 2 < 0$$

$$6x < -2$$

$$x < -\frac{1}{3} \quad \checkmark$$

Question 3



✓ shape

✓ x-ints

✓ TP when $x=1$ (3)

[3]

Question 4

4.1.1. $\pi x^2 + xy = 20$ ✓A

$$\therefore y = \frac{20 - \pi x^2}{x}$$

$$P = 2x + 2y + 2\pi x \quad \checkmark P$$

$$= 2x + 2\pi x + 2 \left(\frac{20 - \pi x^2}{x} \right) \quad \checkmark$$
$$= 2x + 2\pi x + \frac{40}{x} - 2\pi x$$

$$= 2x + \frac{40}{x} \quad (3)$$

4.1.2. $P = 2x + 40x^{-1}$

min when:

$$P' = 2 - 40x^{-2} = 0 \quad \checkmark P' \quad \checkmark P' = 0$$

$$x^{-2} = 2$$

$$x^{-2} = \frac{1}{20}$$

$$x = \pm \sqrt{20} = \pm 2\sqrt{5} \quad \checkmark$$

$$\therefore \min P = 2(2\sqrt{5}) + \frac{40}{2\sqrt{5}}$$

$$\approx 17,89 \text{ cm} \quad \checkmark$$

(4)

4.2.1. $s'(1) = -3(1)^2 + 12(1) = 9 \checkmark$

$$s''(t) = -6t + 12$$

$$s''(1) = -6(1) + 12 = 6 \checkmark$$

$\therefore$ Same sign $\therefore$ speeding up (2)

4.2.2. max when: $-6t + 12 = 0 \checkmark m$

$$-6t = -12$$

$$t = 2 \checkmark A$$

$\therefore$ max speed : $s'(2) = -3(2)^2 + 12(2)$
 $= 12 m/s \checkmark$ (3)

Alternate:

max when : $t = \frac{-12}{2(-3)} \checkmark m$

$$t = 2 \checkmark$$

$\therefore$ max speed : $s'(2) = 12 m/s \checkmark$ (3)

4.2.3. $-3t^2 + 12t = 0$ ✓ $s'(t) = 0$ at A & B

$$t^2 - 4t = 0$$

$$t(t - 4) = 0$$

$$t = 0 \quad \text{or} \quad t = 4 \quad \checkmark$$

$$s(t) = -t^3 + 6t^2 \quad \checkmark s(t)$$

$$\therefore s(4) = -(4)^3 + 6(4)^2$$

$$= 32 \text{ metres} \quad \checkmark$$

(4)

[16]